

Benchmark for Crack Re-Identification in Ticket Management Systems

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Abstract—Building ticket management systems must determine whether a newly reported defect corresponds to an existing ticket or requires a new one. Repeated reports of the same physical incident, such as a crack in a wall photographed from different viewpoints, can generate multiple tickets, fragmenting the incident history and duplicating inspection and repair work. Reliable visual association is therefore essential for linking reports to the correct ticket. We study this association problem through crack re-identification and introduce CRACKID, a dataset containing 140 photographs of 17 walls and 212 crack identities. We evaluate ten zero-shot crop-based matchers from four method families using retrieval, thresholded identity decisions, and open-set identification, with explicit reference levels. Although retrieval performance varies across methods, thresholded same-crack decisions remain close to the best-constant-predictor baseline. Context and crack-erasure controls indicate that wall location contributes more strongly than local crack appearance. A geometric reference baseline registers photographs using wall texture and matches projected crack masks, achieving an open-set identification rate of 0.768 at a false-alarm rate of 0.1 while scoring 48% of query-gallery pairs (prior $n_Q=280$ snapshot; crop rows are the $n_Q=306$ rerun and are not directly comparable). Its limited coverage, failures on smooth walls, and shared annotation provenance constrain this result. These findings establish a dataset and benchmark task for the visual association step needed to consolidate duplicate defect tickets, while leaving real cross-visit performance and end-to-end ticket reduction for future evaluation.

Index Terms—crack re-identification, dataset, benchmark, building maintenance, structural inspection, homography, geometric alignment, annotation quality

I. INTRODUCTION

Building ticket management systems organise reports of physical defects into records that support inspection, prioritisation, intervention, and follow-up. A ticket should maintain the history of a particular incident, bringing together its observations and the actions taken to address it. However, the same damage may be reported repeatedly by different users or during successive inspections. For example, several photographs of a crack in the same wall may lead to separate tickets, even though they refer to a single physical defect. This redundancy can fragment the incident history, duplicate assessment work, and make it difficult to determine whether an intervention remains pending or whether the damage has progressed.

Therefore, addressing this problem requires associating each newly reported observation with the appropriate existing ticket, while allowing a new ticket to be created when the observation concerns a previously unrecorded defect.

Photographic evidence makes this association a visual re-identification problem: the system must determine whether two observations represent the same physical damage. Two distinct scenarios are common in building inspections. In the first scenario, the damage retains essentially the same physical morphology as in previous observations, but the photographs differ in viewpoint, camera distance, field of view, illumination, resolution, and partial visibility, which can substantially alter the recorded appearance of an unchanged crack. In the second scenario, the damage already has a registered ticket but has physically evolved since the previous observation. A crack may extend, widen, branch, or develop additional connected segments, changing its morphology, geometry and photometrics. Here, re-identification must preserve the identity of the incident while recognising that its condition has changed. This scenario requires distinguishing changes in the damage itself from changes caused by photographic acquisition. In both scenarios, an incorrect association may either create a duplicate ticket for the progressing defect or attach the observation to a different incident.

Existing crack-analysis methods primarily address detection and segmentation: they identify the presence of cracks and delineate their extent within individual images. Datasets such as CrackSeg9k [1] and OmniCrack30k [2] support these tasks but do not provide cross-observation identity labels. Consequently, successful detection does not establish whether an observed crack corresponds to one already documented in a ticket. Appearance-based matching approaches, including those developed for person and object re-identification [3], [4], offer candidate methods for this association. Their applicability to cracks nevertheless requires evaluation, particularly when different defects have similar local appearances or when the same defect is observed under changing acquisition conditions or physical progression.

We introduce CRACKID, a dataset for evaluating whether different photographic observations correspond to the same wall crack, supporting the visual association required by building ticket management. The real dataset addresses changes in acquisition conditions within a single inspection visit. A complementary synthetic protocol provides a preliminary assessment of appearance variation and damage progression; performance across real successive inspections remains to be established.

Our benchmark compares ten matching methods without training them on CRACKID. Evaluation distinguishes retriev-

ing the correct candidate from deciding whether two observations represent the same crack, and includes cases in which no corresponding record exists. This distinction is essential for ticket management: placing an existing record near the top of a candidate list is insufficient to justify attaching a new report to it. Experiments with surrounding wall context and crack-erased images further examine whether matching depends on the crack itself or on its location. A geometric reference baseline makes location explicit by registering photographs using wall texture and comparing projected crack masks. The results motivate the use of spatial context alongside local appearance, while registration failures and shared annotation provenance limit the conclusions that can be drawn.

The study is guided by three research questions. RQ1: Can the evaluated methods associate observations of the same defect under varying image acquisition conditions? RQ2: How do wall context and geometric alignment contribute to defect re-identification? RQ3: How do two selected approaches perform under synthetic changes in appearance and crack growth? The first two questions are investigated using real photographs acquired within a single inspection visit. The third provides an exploratory assessment under controlled synthetic transformations. Together, these questions examine the visual association component needed to link new reports to existing defect tickets, while leaving performance across real successive inspections and the reduction of duplicate tickets for future evaluation.

Contributions: (i) the CRACKID crack re-identification dataset with cross-observation identity labels and an explicit account of annotation provenance (Sec. III-B); (ii) a benchmark evaluation of ten matching methods that separates candidate retrieval, same-defect decisions, and identification when no corresponding record exists (Sec. IV, VI); (iii) context controls and a geometric reference baseline for examining the roles of crack appearance, wall location, and registration coverage (Sec. V, VI-C, VII-B); (iv) a preliminary synthetic revisit evaluation that explores appearance variation and damage progression beyond the real dataset’s within-visit scope (Sec. VII-D).

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the CRACKID dataset acquisition and annotation, and Section IV defines the benchmark protocol and evaluation metrics. Section V presents the geometric reference baseline. Section VI reports the crop-baseline benchmark and the wall-context controls (the appearance ceiling). Section VII compares geometry against appearance, analyses registration coverage and failures, ablates the geometric design, and reports the synthetic revisit experiments. Section VIII discusses validity, limitations, and future research, including real cross-visit data and integration into ticket lifecycle evaluation. Section IX concludes the paper.

II. RELATED WORK

Crack analysis is overwhelmingly focused on detection and segmentation [5]–[7]; its datasets annotate cracks within

single images and the large consolidations merging [1], [2] inherit that scope, and multi-view crack pipelines fuse several photographs of one inspection into a surface or a segmentation [8] without solving instance correspondence between them. Structural-health monitoring does compare cracks across time with machinery close to ours [9]–[12], but identity is always *given by construction* (fixed camera, marked location, operator at a viewpoint): the subject is one known crack, the question of width or extent, and no dataset supports method comparison on the association decision itself. Those systems therefore solve measurement after correspondence has already been supplied; they do not decide whether a newly observed crack is the same physical defect as a gallery crack under an open-set protocol. The benchmark targets that missing association decision, while its real capture scope remains within one walk-around rather than a true longitudinal study.

Object re-identification [4] with open-set identification [13] is the formal setting we adopt; keypoint matching [14], [15], learned correspondence (SuperGlue [16], LoFTR [17]), and region embeddings [3], [18]–[21] apply directly to crack crops, and our benchmark measures all of them zero-shot, without CRACKID fine-tuning. We cite LightGlue [22] and place recognition (NetVLAD [23]) as the natural retrieval front end, but we benchmark neither given compute and scope limits and the within-visit focus; they remain future baselines, not claimed comparisons. The YOLOv8 row uses the `yolov8n.pt` detector backbone (layer 9) as an embedding [24], version-pinned in code, not a fine-tuned crack model. It is evaluated zero-shot, without CRACKID fine-tuning. Vehicle-damage assessment already re-identifies damage across claims in curated settings [25], a task the open-set protocol here mirrors; the rest of that literature treats damage detection and severity rating rather than association [26]–[28]. Whether the crop scope is itself the limiting choice for cracks is the question the benchmark is built to answer. The move the baseline makes, aligning two captures and then reasoning about what is present in one and absent in the other, is change detection [29], [30]; planar registration is classical [31]–[33] with photometric alignment (ECC [34]) as the fallback. The contribution here is not a new estimator but the observation that the alignment target and the matching target should be *disjoint*: the homography is driven by the wall precisely because the crack is what is to be identified.

III. DATA ACQUISITION

A. Acquisition and composition

The CRACKID dataset snapshot contains 140 photographs of 17 exterior walls acquired at one site over four half-days with two consumer handsets, at 3072×4080 . The dataset snapshot is archived at Mendeley Data (<https://data.mendeley.com/>). The walls span two surface families that behave very differently under keypoint matching: stippled render (densely textured) and smooth painted plaster, where the crack is frequently the only structure in frame. Walls split disjointly into 6 validation walls (66 photographs, 66 identities) and 11 test walls (74 photographs, 146 identities). Statistical power

is governed by identities visible in two or more photographs, since only these form a query with an answer: 18 in validation, 19 in test. Each wall was captured in one continuous walk-around (consecutive frames a median of 3 s apart), so the dataset measures viewpoint and scale correspondence within one visit, a hypothesized upper bound on multi-visit performance, not yet a measurement of it.

Adjacent-frame retrieval. On the prior snapshot, 100% of 280 answerable test queries had their nearest correct gallery entry in the adjacent frame; on the current labels 95.8% of 306 do. Ranking therefore rewards short-baseline correspondence; Sec. VII sweeps the exclusion gap (prior pool 280→205→149→105; current pool 306→247→188→124 with floors 0.337→0.288→0.250→0.204) and reports that Rank-1 stays flat while the pool and prevalence change with it, so the sweep characterises robustness, not proof that duplicates are irrelevant.

B. Annotation, and why it needs auditing

One click point per crack is propagated across consecutive photographs through an ORB homography and one-to-one assignment; identities whose components appear to be fragments of one physical crack are then merged. The merge is where risk concentrates: a wrongly joined merge converts a hard negative into a free positive and inflates *every* method’s mAP together, so it cannot be caught by comparing methods. We therefore report its extent rather than assume its correctness: 95/212 identities (45%) were formed by a merge (48% on test), bridging a median maximum gap of 228 px; the adjusted Rand index against the pre-merge labeling is 0.080, meaning the merge restructured the partition rather than refining it. All annotations were validated by a human: a single annotator placed and reviewed all 2,095 clicks from homography-propagated prefill until zero provisional points remained. The triage is complete: every one of the 1,023 clicks no component claimed was reviewed against the imagery — 916 sit on visible cracks the segmenter missed rather than labelling mistakes, leaving 107 clicks (10.5%, 95% CI [8.7%, 12.5%]) naming no crack (dataset/labels/_unresolved.json). The 2,095 clicks resolve to 1,051 connected components; “points” below means clicks, not components. Single-annotator; agreement unmeasured, so the merge risk above is bounded by reporting, not removed. Counts in this section and Table I refer to the frozen benchmark snapshot (legacy directory `benchmark_out/`, historic spelling retained for reproducibility); `dataset/labels/` has since been re-edited (walls 18–19 added, audit in progress) and its raw identity/point totals no longer match exactly. The snapshot, its `result.txt`, `chance.json`, `pair_outcomes.json`, and `viewpoint.json` are the version of record for all accuracy rows.

IV. BENCHMARK PROTOCOL

The benchmark evaluates the visual association component required by building ticket management: given a newly reported observation of a crack, determine whether it corresponds to a previously recorded defect or has no match-

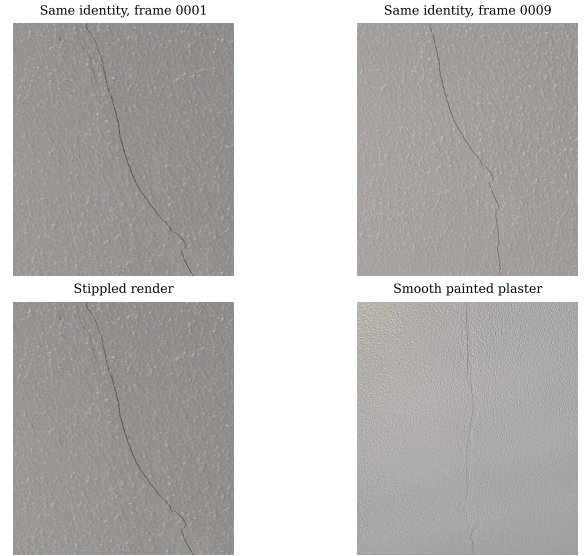


Fig. 1. Examples from real CRACKID captures. Top: the same annotated identity in non-adjacent frames of one walk-around. Bottom: the two surface families, stippled render and smooth painted plaster. The identity labels inherit the dataset’s homography-based propagation and are illustrative, not an independent annotation audit.

ing record in the available reference set. Annotated crack identities serve as experimental proxies for persistent defect records, while photographs provide observations that could be attached to their corresponding tickets. The benchmark measures the accuracy of this association component; it does not directly measure duplicate-ticket reduction or the performance of a complete ticket management workflow. Two evaluation scenarios are considered. The first concerns damage whose physical morphology remains approximately unchanged but whose photographic appearance varies with acquisition conditions. The second concerns previously recorded damage whose appearance or structure changes between observations. Real photographs support the first scenario, whereas controlled synthetic transformations provide a preliminary assessment of the second.

A. Task formulation and candidate records

A *wall* is one physical wall at the capture site. An *identity* is an annotation label linking observations attributed to the same crack across photographs. These labels define the reference correspondence used for evaluation; they are not independent verification of physical identity, and their annotation provenance is described in Section III-B. A *query* is one annotated crack instance to be associated with a reference record. Its *gallery* contains eligible crack instances from other photographs of the same wall, excluding the query photograph. Multiple gallery instances may share an identity, representing several observations of the same recorded defect. Restricting candidates to the same wall assumes that the wall has already been identified; cross-wall search and localisation are outside the benchmark’s scope. A query is *known* when at least one eligible gallery instance has the same annotated identity. It

TABLE I
COUNTING UNITS FOR EVERY METRIC (PRIOR $n_Q=280$ SNAPSHOT;
CURRENT POOL IN TEXT).

	Validation	Test
Walls / photographs / identities	6 / 66 / 66	11 / 74 / 146
Identities in ≥ 2 photographs	18	19
Answerable queries	439	280
Positive query-gallery pairs	26 012	5 722
Valid query-gallery pairs	89 118	31 822
Same-wall image pairs	394	301

is *unknown* when no such instance is available. Unknown therefore means that no matching observation exists in the evaluated gallery; it does not necessarily establish that the physical damage has never been recorded elsewhere. Each method assigns a similarity score to query-gallery pairs. These scores support candidate ranking, thresholded same-identity decisions, and identification with rejection. A *query-gallery pair* is the unit used for pairwise classification, whereas an *image pair* comprises two photographs and is the unit used for registration and correspondence assignment. The gallery is an experimental reference set rather than a chronological ticket database. Accordingly, the benchmark evaluates visual association between observations without claiming to reproduce the temporal arrival of reports or the creation and updating of tickets.

Test contains 306 answerable (known) and 199 unknown queries on the current labels (30 390 valid pairs, 6 160 positive, prevalence 0.203, floor 0.337; `banchmark_out/chance_nq305.json`). Table I retains the prior-pool recount for provenance; its test positives/valid counts (5 722/31 822) are superseded for the crop rerun. On test, registration is attempted for all 301 same-wall image pairs; 255 contain at least one scored crack instance and produce validity-mask cells, while 46 yield no valid query-gallery pair after the 200px size filter (see Sec. VII-B).

Counts in Table I refer to the prior $n_Q=280$ snapshot (test prevalence 0.180); the current test pool is 306 answerable, 30 390 valid, 6 160 positive (prevalence 0.203). The registration pool (301 test pairs) differs from the scored-pair subset (255 pairs with validity-mask cells on the prior snapshot). See Sec. III-B for label drift.

B. Unchanged damage under different acquisition conditions

The primary evaluation uses real photographs acquired during one continuous walk-around per wall. Within this short acquisition period, crack morphology is treated as approximately unchanged. Differences between observations arise mainly from viewpoint, camera distance, scale, field of view, illumination, and partial visibility. This scenario tests whether a method can associate different photographic observations of the same defect despite acquisition variation. It also includes distinct cracks from the same wall, which act as competing candidates, and queries with no matching identity in their

eligible galleries. Validation and test walls are disjoint. Methods use their existing configurations and pretrained weights where applicable, without fine-tuning on CRACKID. Decision thresholds are calibrated on validation walls and then applied to test walls. Shared crack masks and instance extraction provide a common input representation across methods; the extraction rules are described in Section V. Because the photographs come from a single visit, this evaluation establishes within-visit correspondence performance. It does not directly establish robustness across separate inspections or under naturally occurring damage progression.

C. Controlled appearance change and damage progression

A complementary synthetic revisit experiment separates changes in photographic appearance from changes in crack structure. Ten source photographs from five walls are used with one random seed. Synthetic queries are compared with the real reference instances from their respective walls. Both conditions apply the same modest affine transformation, using a scale factor of 1.08, a rotation of 6° , and a tilt parameter of 4° . In the appearance-change condition, crack structure is retained while image appearance is modified through Gaussian blur, noise, and contrast and brightness changes. This condition extends the assessment of acquisition variation under controlled transformations. In the damage-growth condition, a single tip of the largest crack component is extended, modifying its extent while preserving its original identity. This condition provides a limited test of whether an evolved observation can still be associated with the previously recorded defect. It represents one form of simulated crack growth; it does not reproduce the full range of widening, branching, deformation, or environmental changes encountered between real inspections. The synthetic experiment compares OSNet with wall context against HYBRID-SKEL. HYBRID-SKEL combines geometric alignment with skeleton comparison through fallback and blending, and is distinct from the REG+CHAMFER reference used in the primary benchmark. The two experiments therefore evaluate different geometric configurations and should be interpreted separately. With one seed and five source walls, the synthetic results are preliminary point estimates rather than evidence of real longitudinal performance.

D. Decision rules and handling of unavailable scores

Thresholded pairwise classification accepts a query-gallery pair as a correspondence when its similarity score meets the selected threshold. For query-level identification, the highest-scoring gallery instance is accepted only when its score meets that threshold; otherwise, the query is rejected. Correspondence assignment is also evaluated independently within each query-image/gallery-image block. The Hungarian algorithm maximises total similarity under a one-to-one constraint, after which assignments below the validation-selected threshold are rejected. This constraint applies to instances within an image pair. It does not impose a one-to-one relationship between reports and tickets: several observations may belong to the same persistent defect record. Pairs for which a method cannot

produce a finite score are retained in evaluation. They count as predicted negatives for pairwise and assignment classification and rank below scored candidates for retrieval. A known query without a finite candidate score is a missed association. An unknown query without a finite score produces no false association. For interpretation in ticket management, an unavailable score must nevertheless be distinguished from a supported non-match. Registration failure or insufficient visual evidence indicates an unresolved association, which may require review; it does not establish that a new ticket should be created.

For assignment F_1 , correct accepted assignments are true positives, incorrect accepted assignments are false positives, and unmatched known instances are false negatives. Thus $F_1 = 2TP / (2TP + FP + FN)$; unmatched unknown queries are correct rejections. Unscorable pairs count as predicted negative for pairwise and assignment F_1 and rank last for retrieval. A known query with no finite score is a miss, while an unknown query with no finite score is not a false alarm.

E. Evaluation metrics and reference baselines

Candidate retrieval is evaluated over known queries using Rank-1, Rank-5, and mean average precision. These metrics measure whether observations of the correct identity appear near the top of the candidate list.

Pairwise identity decisions are evaluated using precision, recall, and F_1 over valid query-gallery pairs. We distinguish the oracle test threshold, reported as F_1^* , from the validation-calibrated threshold, reported as $F_1^{@v}$. The oracle result describes an optimistic performance reference; the validation-calibrated result evaluates a decision threshold selected without using test labels.

Assignment accuracy, reported as F_1^a , evaluates accepted correspondences within image pairs. Correct accepted assignments are true positives, incorrect accepted assignments are false positives, and unmatched known instances are false negatives.

Identification with rejection is evaluated through detection and identification rate, DIR, and false-alarm rate, FAR. At threshold t , DIR is the proportion of known queries whose highest-scoring candidate is correct and accepted. FAR is the proportion of unknown queries for which a candidate is incorrectly accepted. Incorrect associations of known queries reduce DIR but are not included in the unknown-query FAR.

DIR@FAR= 0.1 summarises performance at a constrained operating point on the test curve. It is distinct from performance at the fixed validation-calibrated threshold and does not guarantee that the same false-alarm rate will hold in deployment.

Coverage is reported as the proportion of valid pairs receiving a finite score. Reporting coverage alongside association accuracy prevents a method that answers only a subset of cases from being interpreted as providing complete support for ticket association.

Metrics are interpreted against appropriate reference baselines, including random-ranking performance and the pairwise F_1 obtained by predicting every pair as a correspondence. The

latter equals $2p/(1+p)$, where p is the positive-pair prevalence, and is a constant-predictor reference rather than a guaranteed lower bound. Counts, prevalence, and reference values must correspond to the same frozen evaluation snapshot. Results obtained from different snapshots are reported separately.

Every metric is reported against its chance level, because two are far from zero (current test pool, `chance_nq305.json`: 200 random matrices): pairwise F_1 has a best-constant-predictor floor $2p/(1+p) = 0.337$ at prevalence 0.203, Rank-5 chance 0.778 ± 0.019 , mAP 0.382; Rank-1 0.336 ± 0.025 and DIR@FAR 0.038 ± 0.014 , assignment- F_1 0.345. Prior-pool values were 0.305/0.767/0.336/0.038. We call $2p/(1+p)$ a floor, not coin-flip chance: it is the score of predicting every pair positive, and it changes when the evaluated pair pool changes. Confidence resampling uses walls rather than queries because queries from one wall share capture conditions. With 11 test walls, no numeric interval is attached; wall-level intervals await a larger dataset and release of the per-pair score matrices (Sec. VIII).

V. A GEOMETRIC REFERENCE BASELINE

The geometric reference baseline investigates whether the spatial location of a crack on its wall can support association with an existing defect record. Different photographs may depict the same damage with substantial changes in viewpoint, scale, and visible context. By aligning the wall surfaces, the method places these observations in a common coordinate frame and compares their crack masks at corresponding locations. In building ticket management, this provides a candidate mechanism for linking a new observation to a previously documented defect.

The baseline assumes an approximately planar wall surface for which image registration provides a useful spatial correspondence. It combines wall registration, crack-mask projection, and geometric comparison, without training a matching model on CRACKID. Crack masks are produced by the shared UNet++ segmenter (`segmentation-models-pytorch`, `batch_segment.py` / `tiled_inference.py`) and converted into connected-component instances using a minimum area of 200 pixels and morphological closing of 5 pixels. The same segmentation and instance extraction are used across the evaluated methods, keeping the input representation consistent.

A. Registration using wall texture

For photographs I_a and I_b , registration estimates a transformation H that maps points from I_a into the coordinate frame of I_b : $x' \sim Hx$. A homography models the viewpoint change under the approximately planar-surface assumption. The primary registration stage estimates this transformation from the surrounding wall texture while excluding detected cracks. Crack masks are dilated by 15 pixels before feature detection, reducing the contribution of crack pixels and their immediate boundaries. This separation is intended to estimate alignment from surface features independently of the crack correspondence being evaluated. It reduces the risk that similar crack structures drive an alignment that subsequently appears

to confirm their identity. This independence is conditional on the masks and the registration stage; it is not guaranteed for every image pair. Registration operates on image copies downsampled to a maximum long-side dimension of 1600 pixels. Contrast is adjusted using CLAHE with a clip limit of 2.0 and an 8×8 tile grid. SIFT extracts up to 8000 features, and FLANN matches descriptors using a Lowe ratio threshold of 0.75. Robust transformation estimation uses USAC MAGSAC where available, with RANSAC as a fallback, a 4-pixel re-projection threshold, and up to 10 000 iterations. Acceptance requires at least 25 inliers.

B. Registration cascade and validity checks

The available alignment evidence depends on the wall surface. Textured render can provide features outside the cracks, whereas smooth painted plaster may contain too little background structure for reliable matching. The baseline therefore uses a three-stage registration cascade. The first stage applies SIFT with crack regions excluded. If masking leaves fewer than 300 keypoints in either image, a second SIFT stage restores the crack regions. This fallback increases the available feature set, but allows the damage itself to contribute to alignment. Its results must therefore be distinguished from those obtained using masked wall texture. The third stage uses ECC photometric alignment over a $320 \rightarrow 640 \rightarrow 1280$ -pixel pyramid, with translation followed by Euclidean initialisation and up to 300 iterations per level. An ECC result is accepted only when the enhanced correlation exceeds 0.80. Candidate transformations are checked for geometric plausibility. Mirrored or folded quadrilaterals, collapsed sides, and area scale factors outside $[1/16, 16]$ are rejected. The successful stage—masked SIFT, unmasked SIFT, or ECC—is recorded for each image pair (`sift`, `sift-nocrackmask`, `ecc`). When available, the full-resolution inlier root-mean-square error provides an estimate of registration uncertainty used in the mask-comparison tolerance. If no stage produces an accepted transformation, the image pair remains unscorable. Registration failure indicates insufficient alignment evidence and does not establish that the observations represent different defects.

C. Crack projection and geometric comparison

Following accepted registration, crack-instance pixels from I_a are projected into the coordinate frame of I_b . Each projected mask is compared with candidate instances in the reference photograph. The primary comparison uses symmetric mean Chamfer distance. This measure is less sensitive to small alignment errors than direct mask overlap: because cracks are narrow, a small displacement can substantially reduce intersection-over-union even when two masks depict the same crack. Let a be a projected crack mask, b a candidate reference mask, and $d(p, b) = \min_{q \in b} \|p - q\|_2$ the Euclidean distance

from pixel p to the nearest pixel in b . The symmetric mean Chamfer distance is

$$D(a, b) = \frac{1}{2} \left[\frac{1}{|a|} \sum_{p \in a} d(p, b) + \frac{1}{|b|} \sum_{q \in b} d(q, a) \right], \quad (1)$$

over the union bounding box, with early rejection of spatially distant candidates. Distances are measured in full-resolution pixels, with smaller values indicating closer geometric agreement.

As a complementary measure, Chamfer coverage evaluates the fraction of mask pixels lying within tolerance τ of the other mask: $\frac{1}{2}(|\{p \in a : d(p, b) \leq \tau\}|/|a| + |\{q \in b : d(q, a) \leq \tau\}|/|b|)$. The tolerance is derived from each image pair’s registration error and clamped to $[2, 16]$ pixels. A default of 6 pixels is used when that error is unavailable (the synthetic HYBRID-SKEL variant fixes $\tau=8.0$ px). Coverage expresses agreement as a proportion, although it remains dependent on the selected spatial tolerance. The two measures are evaluated separately to examine how the comparison rule affects association performance (Sec. VII-C). Eq. (1) is a per-pair pixel distance whose scale differs per pair, so a single global threshold makes ranking partly measure calibration.

D. Correspondence decisions and limits of spatial evidence

Correspondences are selected using the Hungarian algorithm [35] and a validation-calibrated threshold. One-to-one assignment applies within each image pair; multiple observations may belong to the same defect record. The Hungarian solve is part of *evaluation*, applied identically to every method’s score matrix. After alignment, geometric agreement supports a match. Missing detections provide only conditional evidence of a non-match, while locations outside the field of view and registration failures remain unresolved. With I_a projected into I_b ’s frame, the system observes a region visible and empty of the recorded crack—a **licensed negative**, positive evidence of absence, categorically stronger than the low similarity a crop embedding produces. Unscorable pairs (no registration, or no retained instance after the 200 px size filter) receive $-\infty$ and are charged as predicted negative for F_1 and rank last for retrieval. REG+CHAMFER primarily addresses unchanged damage under varying acquisition conditions. Damage progression is explored separately using HYBRID-SKEL. Results must also account for surface-dependent registration failures and the dataset’s use of homography-based annotation.

VI. THE APPEARANCE CEILING

A. Setup

The benchmark compares approaches to the visual association of crack observations with existing defect records. Ten methods operate on crack-centred image crops, while REG+CHAMFER uses registration of the wall photographs and geometric comparison of projected crack masks. Additional experiments examine the contribution of surrounding wall context and evaluate a separate geometric configuration under synthetic appearance change and damage growth.

TABLE II

MAIN RESULT ON CRACKID TEST ($n_Q=306$ OVER 11 WALLS). F_1^* IS ORACLE-THRESHOLDED ON TEST; ΔF_1^* IS EXCESS OVER THE 0.337 FLOOR; $F_1^{\textcircled{v}}$ USES THE VALIDATION-CALIBRATED THRESHOLD. BOLD: BEST NEW CROP ROW PER COLUMN. [†]CLIP, COVERAGE-ONLY, AND REG+CHAMFER ROWS ARE THE PRIOR $n_Q=280$ SNAPSHOT, NOT COMPARABLE TO THE NEW ROWS. SCORING, CHANCE LEVELS, AND COSTS: SEC. IV–VII-B.

Method	Family	R@1	mAP	DIR@FAR=0.1	F_1^*	ΔF_1^*	$F_1^{\textcircled{v}}$	scored	s/query
<i>Reference level (same 0.48 pool as geometry)</i>									
COVERAGE-ONLY	geometry	0.515	0.533	0.000	0.544	0.239	0.544	0.48	—
<i>Crop scope (new $n_Q=306$ rerun except CLIP)</i>									
ORB [15]	keypoint	0.370	0.372	0.043	0.337	0.000	0.337	1.00	0.077
SIFT [14]	keypoint	0.420	0.408	0.049	0.343	0.006	0.337	1.00	0.155
SuperGlue [16]	learned	0.518	0.434	0.384	0.357	0.020	0.337	1.00	5.196
LoFTR [17]	learned	0.512	0.424	0.285	0.357	0.020	0.337	1.00	7.190
YOLOv8 backbone [24]	embedding	0.525	0.439	0.134	0.374	0.037	0.338	1.00	0.016
ViT-B/16 [18]	embedding	0.613	0.487	0.259	0.344	0.007	0.336	1.00	0.021
DeiT-S [19]	embedding	0.570	0.479	0.223	0.337	0.000	0.337	1.00	0.008
CLIP ViT-B/32 [20]	embedding [†]	0.604	0.471	0.143	0.307	0.002	0.306	1.00	0.050
DINOv2 [21]	embedding	0.567	0.459	0.272	0.337	0.000	0.331	1.00	0.011
OSNet [3]	re-ID	0.603	0.497	0.154	0.338	0.001	0.337	1.00	0.006
<i>Full-image scope</i>									
REG+CHAMFER (reference)	geometry	0.952	0.886	0.768	0.656	0.351	0.582	0.48	27.92

1) *Local feature matching*: SIFT [14] and ORB [15] represent traditional local-feature approaches. They detect keypoints and compare descriptors within crack crops. Candidate correspondences are geometrically verified using RANSAC, and the number of inlier correspondences provides the matching score under the same validity mask. SuperGlue [16] and LoFTR [17] represent learned correspondence approaches. Both use their pretrained indoor configurations, with inputs resized to a maximum long-side dimension of 512 pixels. Their correspondences are evaluated through the same geometric verification and validity rules used for the traditional methods. These approaches test whether repeatable local structures can identify a crack across different photographic observations.

2) *Image representations and re-identification*: Five methods use general visual representations: ViT-B/16 [18], DeiT-S [19], CLIP ViT-B/32 [20], DINOv2 ViT-S/14 [21], and the YOLOv8 backbone [24]. CLIP uses the `openai/clip-vit-base-patch32` configuration, while YOLOv8 features are extracted from layer 9 of `yolov8n.pt`. OSNet [3], using the `osnet_x1_0` configuration, represents an architecture developed specifically for re-identification. It is evaluated with its existing pretrained weights, without adaptation to crack identities. Embedding inputs are letterboxed to 224×224 pixels with ImageNet normalisation. Cosine similarity between feature vectors provides the matching score. These methods assess whether compact visual representations can distinguish crack identities despite acquisition variation. All backbones are zero-shot on CRACKID; no backbone is fine-tuned on cracks.

3) *Wall context and crack-erasure controls*: Context experiments use OSNet and CLIP to examine whether matching depends on the crack appearance or on the surrounding wall. The crop-plus-context condition (`@ctx1`) expands the crack-instance bounding box by one original box width and height

on each side. The crack-erased condition (`@ctx1@nocrack`) uses the same expanded region but removes the detected crack through Telea inpainting, with a 5-pixel mask dilation and a 3-pixel inpainting radius, so the silhouette is removed rather than blacked out. The backbone and scoring procedure remain fixed across these conditions. Comparing tight crops, expanded crops, and crack-erased crops isolates the contribution of the additional visual context.

4) *Geometric approaches and coverage control*: REG+CHAMFER follows the registration and mask-comparison procedure described in Section V. It uses spatial correspondence between wall photographs to compare crack instances in a common coordinate frame. A COVERAGE-ONLY control assigns one constant score to pairs from successfully registered images and no finite score to unregistered pairs. It does not compare crack shape or appearance. This control measures how much performance can arise from the pattern of registration success alone. The synthetic experiment compares OSNet with context against HYBRID-SKEL. HYBRID-SKEL supplements geometric alignment with skeleton comparison through fallback and blending, using a fixed tolerance of 8 pixels. It is a separate configuration from REG+CHAMFER. LightGlue [22] and NetVLAD [23] are relevant candidates for future comparison but are not evaluated in this benchmark given compute and scope limits.

B. Result: appearance buys almost nothing on the decision

Table II summarises the primary benchmark for unchanged damage under acquisition variation. Among the nine updated crop methods, Rank-1 ranges from 0.370 to 0.613. ViT-B/16 achieves the highest Rank-1, while OSNet achieves the highest mAP. SuperGlue obtains the highest DIR@FAR= 0.1 among these methods.

On the decision a record actually makes—same crack or not, at a threshold—no new crop row clears the best-constant

floor by more than 0.037. At the oracle threshold on test, new F_1^* spans 0.337 to 0.374 against the 0.337 floor at prevalence 0.203 (Table II), i.e. +0.000 to +0.037, with three rows at the floor to three decimals. This band spans handcrafted keypoints to foundation-model embeddings and more than two orders of relative compute (0.006 s to 7.19 s per query in separate per-method runs, Table II; absolute timings are workstation-dependent).

At a deployable validation-calibrated threshold, the nine new rows sit at $F_1^{\text{val}}=0.331\text{--}0.338$ against the 0.337 floor (−0.006 to +0.001): thresholded decisions are at or below chance.

Ranking does not transfer to deciding. the nine new rows separate over a 0.243 range on Rank-1 (0.370–0.613; chance 0.336 ± 0.025), yet ViT takes best Rank-1 (0.613) while its oracle pairwise F_1^* sits only 0.007 above the floor, and SuperGlue (seventh on Rank-1) has the best new crop open-set rate (0.384; chance 0.038 ± 0.014). Ordering a gallery well and keeping a record are different capabilities. The collapse after validation calibration is consistent with threshold instability under a small, wall-clustered sample: methods can order candidates differently while their score distributions do not support a stable same-crack/not-same-crack boundary on held-out walls. This is an interpretation of the observed pattern, not a claim that calibration itself is the sole cause. With 19 matchable test identities and no wall-level intervals, no fine ranking among crop baselines is claimed.

One contributor is checkable from the counting units. Validation pair prevalence was 0.292 (26 012/89 118) on the prior snapshot against 0.203 on the current test pool (6 160/30 390), so the F_1 -optimal threshold is not a property of a method but of the pool it was fitted on: the same scores that separate cracks on the denser validation pool place their boundary in the wrong place once prevalence shifts, and every method shifts together. The current test floor is 0.337 (validation floor TBD after val-split rerun with walls 18–19). This is a mechanism, not a measurement: the threshold-free version of the claim (pairwise average precision, or AUROC against prevalence) would settle it, and we do not report it; releasing the saved score matrices it requires is listed as future work (Sec. VIII).

C. Is it the crack, or the wall behind it?

In the earlier snapshot, adding context increases OSNet’s Rank-1 from 0.654 to 0.775 and DIR@FAR= 0.1 from 0.168 to 0.404 (prior $n_Q=280$ snapshot; context controls pending rerun at $n_Q=306$). Two controls separate identity cue from field of view at fixed backbone: a *crop+context* condition (@ctx1: instance bounding box expanded by $1.0\times$ its size on each side) and a *crack-erased* condition (@ctx1@nocrack: the same widened view with the crack mask dilated by 5 px and filled by Telea inpainting, radius 3 px, so the silhouette is removed rather than blacked out). Context helps a lot: OSNet’s Rank-1 rises from 0.654 to 0.775, mAP to 0.595, open-set rate from 0.168 to 0.404, and its pairwise F_1^* from 0.309 to 0.354 (prior $n_Q=280$ snapshot; context controls pending rerun at $n_Q=306$); CLIP’s open-set rate rises from 0.143 to 0.311, but

erasing the crack costs almost none of it: OSNet retains Rank-1 0.711, mAP 0.574 and open-set 0.393; CLIP retains Rank-1 0.600, mAP 0.502, open-set 0.286. Both erased views still outrank every tight-crop method: the retrieved cue is the wall, that is, location. The persistence of performance after crack erasure indicates that surrounding wall cues contribute strongly to association in this dataset. These cues may help recognise the location of an observation even when the crack itself provides limited distinguishing information. The experiment supports the use of context, while leaving its robustness across separate inspections to be established. RQ2 (context part): wall context improves retrieval and identification with rejection. Much of this benefit persists after crack erasure, indicating that surrounding wall cues contribute substantially to association in this dataset. Pairwise F_1 moves little with context on the prior pool: all four context variants lie between 0.346 and 0.361 (prior floor 0.305), over a change that gives OSNet +0.236 of open-set rate. We read the aggregate as a property of the data, pending rerun at the current pool: methods with no shared inductive bias land near the floor, and wherever a crop method improves, the improvement is not the crack. Ten methods are a sample, not proof; the result establishes that the obvious moves buy at most 0.037 above the 0.337 floor on the decision under the new rerun, reason to change the formulation, not the encoder. Differences in retrieval performance are accompanied by a much narrower range of thresholded pairwise accuracy, so a method’s ability to place the correct candidate near the top of a list does not necessarily translate into a reliable same-defect decision. For ticket management, a ranked list may help an operator locate a related record, while accepting that association requires additional evidence. Therefore (RQ1): although methods can retrieve observations of the same crack under within-visit acquisition variation, reliable automatic identity decisions remain limited.

VII. RESULTS AND DISCUSSION

Results are organised around the association decisions required by building ticket management: retrieving the appropriate existing record, accepting a correct association, and rejecting an observation without a matching reference. Registration coverage and computational cost are considered alongside accuracy. The updated crop-method evaluation uses 306 answerable queries (current labels; `chance_nq305.json`). CLIP, context controls, and the geometric experiments currently retain results from the earlier 280-query snapshot. These groups are reported separately because their evaluation populations and reference values differ. Results are point estimates, and the limited number of matchable test identities does not support confident fine-grained ranking between methods.

A. Main comparison and the open-set gap

The geometric reference (prior $n_Q=280$ snapshot) and the new crop rerun ($n_Q=306$) are on different pair pools and are not numerically comparable. Prior crop Rank-1 peaks at 0.654 against geometric Rank-1 0.952; crop mAP peaks at 0.520 against 0.886, but the pools differ (*scored* 1.00 vs 0.48).

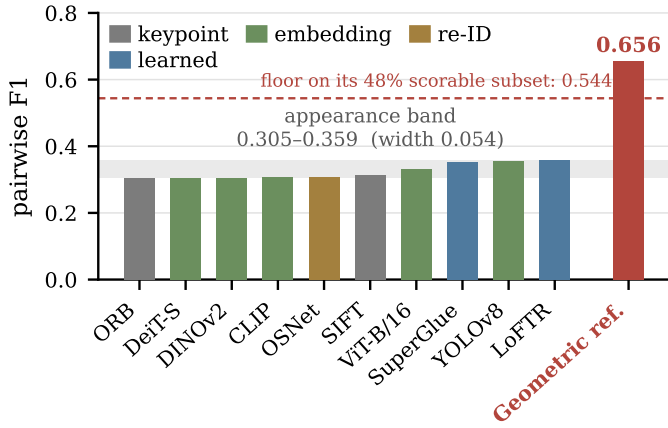


Fig. 2. The appearance ceiling. Pairwise F_1 for ten crop-scope matchers spanning four method families on CRACKID test ($n_Q=306$ for nine rows; CLIP-crop prior snapshot). New rows fall inside 0.337 to 0.374 against the 0.337 best-constant floor ($2p/(1+p)$ at prevalence 0.203), irrespective of family or relative compute. The geometric reference baseline (red, prior snapshot) leaves the band, but it answers only 48% of pairs, for which the best-constant floor on the current labels is 0.558 (the COVERAGE-ONLY level with old registration outcomes), not 0.337: its bar is drawn against the dashed line and must not be read against the crop floor. Ranking metrics separate the same methods over a 0.24 range; the decision metric does not. All bars are point estimates on the rerun pool with no wall-level intervals. Figure bars show the prior-pool band and await regeneration after the full-pool rerun; numbers in Table II take precedence.

Conditioned on coverage, as excess over the COVERAGE-ONLY reference, the prior ranking gains are $+0.437/+0.353$, while prior pairwise $F_1^* = 0.656$ is $+0.112$ over 0.544, not the $+0.351$ a comparison against the crop floor would suggest. The open-set result is the one number coverage does not explain: prior $\text{DIR@FAR}=0.1$ reaches 0.768 against 0.384 for the best new crop method (SuperGlue at $n_Q=306$), 0.404 for the best prior context control, chance 0.038 ± 0.014 , and exactly 0.000 for COVERAGE-ONLY. Observed: crop methods return a similarity to the nearest neighbor, which on self-similar hairline cracks is often high, so they cannot decline. Interpretation: an aligned pair supports the *licensed negative*: the query projects into the gallery frame, where its region is visible and empty. Relative compute does not explain the gap: LoFTR is the costliest new crop row (7.19 s/query), OSNet the cheapest (0.006 s/query) in separate per-method runs (crop costs: $\text{total_s}/306$; absolute timings workstation-dependent). RQ2 (geometric part): geometric alignment enables crack comparison at corresponding wall locations and provides identification information beyond registration coverage alone. Its contribution is conditional on successful alignment, surface characteristics, and annotation provenance; results from different benchmark snapshots do not establish a direct ranking against all crop methods.

B. Partial coverage, gap structure, and failures

Every crop method scores 100% of valid pairs, the geometric method 48%; Unscorable pairs are charged as predicted negatives. For ranking that is a penalty: censoring a complete matrix drops Rank-1 0.422 to 0.27 at 34% coverage in a

held-out control (score matrices not committed; see paper README). For pairwise F_1 it is a subsidy: the scorable subset retains 93.0% of positives and 37.5% of negatives on the current labels with old registration outcomes (100%/36.8% prior), so prevalence rises 0.203 to 0.387 and the floor with it, 0.337 to 0.558 (prior: 0.180 $\rightarrow$ 0.373, 0.305 $\rightarrow$ 0.544). COVERAGE-ONLY scores 0.544 pairwise F_1 on the prior pool (0.558 on current labels with old outcomes) and mAP 0.533 on the prior pool (both above prior OSNet 0.309/0.520; current-pool rerun of coverage/geometry pending), and Rank-1 0.515 prior-pool, by answering every registered pair without examining a crack. Its signal is registration overlap: it assigns one constant score to registered image pairs and $-\infty$ to unregistered pairs, so candidate images are ranked even though crack pixels are never inspected. Much of the ranking and pairwise- F_1 margin is therefore *which pairs align*; the open-set metric is the exception and the result we lead with. Successful registration selects a non-uniform subset of the evaluation population. REG+CHAMFER exceeds this control, including a DIR of 0.768 compared with 0.000. Within this snapshot, geometric comparison contributes identification information beyond registration availability. Nevertheless, its performance remains conditional on limited coverage and on annotation provenance shared with the registration-based approach. Failures are strongly wall-dependent, with no successful registration on two test walls. Consequently, the aggregate result cannot be interpreted as uniform support across building surfaces.

Excluding the adjacent frame, then the two and three nearest, leaves Rank-1 flat on the prior pool (OSNet 0.654 to 0.648) with excess over chance stable in 0.24 to 0.32: the near-duplicate structure is real but the flat curve alone does not prove it is irrelevant, because the answerable pool shrinks (prior 280 $\rightarrow$ 105; current 306 $\rightarrow$ 124) and prevalence shifts with it. It is also important for interpretation: the adjacent frame answers almost all queries (100% prior; 95.8% current), so the headline ranking measures short-baseline correspondence rather than broad temporal re-identification. Registration success is strongly wall-dependent (97/301 pairs, 32.2%: walls 07/08/12 100%, wall 09 52%, wall 11 83%, walls 13-16 between 3% and 29%, walls 10/17 at 0%). The pairs that register are the densely annotated ones, so the 96 registered pairs that carry a scored query-gallery pair hold 48% of the 31 822 valid ones — 160 valid cells per image pair against 104 for a failed pair — from a third of the available pairs (46 of the 301 same-wall pairs yield no valid pair at all). A logistic fit over all 301 pairs makes the mechanism plain: sharpness dominates (median 133.8 registered vs 7.2 failed, wall-clustered Wald $z = 5.1$, $p = 3 \times 10^{-7}$), frame gap far less ($z = -2.8$, $p = 0.005$). The independent-pairs z s (8.2, -5.0) overstate the contrast because failure is categorical by wall, so we resample walls, not queries, and the leave-one-wall-out spread in Rank-1 is large, so no fine ranking among baselines is claimed. One admission gate (variance-of-Laplacian ≥ 10) applied identically rejects 21 of 74 test captures and nearly doubles registration, 32.2% to 56.1%. This suggests a possible

role for capture-quality guidance in a ticket management interface. However, the increase concerns a filtered population and should not be interpreted as improved coverage of all submitted photographs. Excluding adjacent frames and then progressively more neighbouring frames leaves OSNet Rank-1 approximately stable on the prior pool; this result describes performance on a changing candidate population and does not demonstrate robustness across separate inspection visits.

C. Ablations and cost

Chamfer distance versus coverage. The scale-free coverage score motivated in Sec. V is not a free improvement: it drops mAP 0.886 to 0.646 and Rank-1 0.952 to 0.849, the largest movement any change produces; DIR and $F_1^{@v}$ barely move (0.768→0.754, 0.582→0.556), so the choice of geometric comparison affects ranking more strongly than the reported thresholded decisions. Masking. Disabling crack exclusion leaves ranking and pairwise agreement practically unchanged (to three decimals) but raises DIR@FAR 0.768 to 0.825: masking is not what buys the reported accuracy. This is expected when the test set contains no deliberate case where crack pixels create a spurious registration: masking buys evidence independence, not observed accuracy on this collection. We retain masking as the primary configuration because it keeps the alignment target disjoint from the identity target, and report the nomask +0.057 openly as a safeguard awaiting a harder stress test. ECC fallback. With ECC enabled (primary): Rank-1 0.952, mAP 0.886, *scored* 0.48; with ECC disabled: Rank-1 0.920, mAP 0.862, *scored* 0.50. Assignment F_1 rises 0.769 to 0.784 when ECC is enabled; open-set DIR is unchanged. ECC was kept versioned after a fix for a configuration in which it had been inert.

Cost. The geometric pass costs 7817s total on test in our sessions (27.9s/query if charged per query, about 26s per same-wall image pair over 301 pairs). That per-query figure is a charging convention, not an intrinsic one: registration is per image *pair*, one homography serves every crack on a wall, and amortized over the 301 test image pairs the same pass runs about 26s/pair, after which any later crack query on the pair pays only the Chamfer pass, not the homography. Absolute timings are workstation- and load-dependent and are reported for relative ordering only; relatively, the pair-level cost is within an order of magnitude of LoFTR, a crop method with none of the open-set benefit. No fine ranking by cost is claimed.

D. A synthetic revisit protocol

The operational setting a ticket management backend serves is a *revisited* crack — one that already has an open issue ticket — so we add a second protocol isolating controlled change. Ten source photographs across five of the walls (walls 01, 02, 04, 05, 09; a single seed, 0) are each synthesized into a set of queries under an identical affine warp ($1.08\times$ scale, 6° rotation, 4° tilt; affine approximation, not a full projective revisit): dataset A additionally grows a single tip of the largest crack component (length

Protocol	Method	n_q	R@1	mAP	DIR@FAR=0.1
A (growth)	HYBRID-SKEL	69	0.275	0.401	0.343
A (growth)	OSNet+ctx	69	0.826	0.647	0.739
B (appearance)	HYBRID-SKEL	82	0.274	0.403	0.342
B (appearance)	OSNet+ctx	82	0.671	0.570	0.663

TABLE III

HYBRID-SKEL VERSUS THE STRONGEST APPEARANCE BASELINE ON THE SYNTHETIC REVISIT PROTOCOL (SEC. VII-D); PAIRWISE AND ASSIGNMENT F_1 AT THE VALIDATION-CALIBRATED THRESHOLD. ONE SEED (0), TEN SOURCE PHOTOGRAPHS ACROSS FIVE WALLS, NO RESAMPLING; ROWS CARRY NO UNCERTAINTY AND ARE AN ACCURACY CHECKPOINT, NOT A STABILITY CLAIM (SEC. VII-D). HYBRID-SKEL IS NOT THE TABLE II REG+CHAMFER CONFIGURATION.

$0.18\times\max(16, W/14)$ px, width 3px, tip farthest from the component centroid, `hybrid_eval.py`), dataset B corrupts appearance while leaving structure untouched (Gaussian blur $\sigma=1.6$, noise $\sigma=14$, contrast/brightness shift); each synthetic query is ranked against the wall’s real instances only. Unlike the Table II REG+CHAMFER system, which is pure registration plus Chamfer scoring with per-pair RMS tolerance, the synthetic-protocol HYBRID-SKEL folds a structural skeleton comparison into the geometric baseline as a fallback plus per-pair blend ($\tau=8.0$ px), scored by the same code as OSNet+ctx (Table III). HYBRID-SKEL and REG+CHAMFER are therefore different systems; the synthetic rows test generalization under controlled change, not the Table II configuration. The warp is deliberately modest (real walk-around tilt reaches p90 75°), so this is a checkpoint on small geometric change with controlled photometric or growth change, not a real revisit.

The geometric side does not transfer its main-benchmark advantage here: OSNet+ctx separates the synthetic queries at R@1 0.826/0.671, the HYBRID-SKEL at 0.275/0.274, with open-set rates 0.739/0.663 against 0.343/0.342. But the split is metric-dependent, and opposite the ranking gap: on identity agreement, the HYBRID-SKEL holds its own, with assignment $F_1^a = 0.646/0.640$ beating OSNet’s 0.631/0.609, and pairwise $F_1^{@v} = 0.569/0.574$ trailing 0.601/0.594 by far less than the ranking margin. Both rate 1.00 scored, and per assignment logs none was decided by the pure-homography branch alone ($H\% = 0.0$, the share of assignment cells whose verdict came straight from the homography with no skeleton fallback or blend, from the HYBRID-SKEL assignment logs rather than Table III): the geometric pipeline answers *which crack is which* on pairs appearance also resolves, but loses the global ranking of a hundred-plus same-wall candidates at one threshold. These rows are a single-seed checkpoint on five walls, not a claim: they show the Table II advantage does not imply multi-visit robustness. The relative ordering depends on whether evaluation concerns ranking a gallery or assigning correspondences within an image pair. (RQ3) These results do not establish robustness to real damage progression. The transformations are controlled, the geometric change is modest, and the sample covers one seed and five walls. Furthermore, HYBRID-SKEL differs from REG+CHAMFER, so this experiment does not directly test the primary geometric

configuration under progression.

VIII. VALIDITY, LIMITATIONS, AND FUTURE WORK

This work establishes a first measurable formulation of within-visit crack re-identification and a geometric reference baseline. The results are point estimates on a small, wall-clustered sample with single-annotator labels and partial registration coverage, so each finding below is paired with the concrete step that would confirm or extend it.

- 1) **Scale to multi-wall, multi-visit data.** Collect more walls, surface families, and annotators to support wall-level confidence intervals and reliable method ranking. Most importantly, add a second capture session at deliberately varied distances and lighting to convert the within-visit measurement into a genuine cross-visit benchmark for ticket management backends.
- 2) **Add provenance-free annotation.** All clicks were human-validated by a single annotator with full triage of unclaimed points, but agreement is unmeasured and the geometric advantage remains provenance-conditioned. A small subset labeled entirely without homography propagation, plus released per-pair score matrices for sensitivity analysis, wall-clustered intervals, and threshold-free AP/AUROC, would quantify how much of the open-set gap is independent of annotation provenance.
- 3) **Close the registration coverage gap.** The baseline answers 48% of pairs and scores zero on smooth-plaster walls. Surface-adaptive registration — learned local features, direct photometric alignment, or sparse depth from consumer devices — would push coverage toward full-set evaluation and further decouple the geometry pipeline from homography-based provenance.
- 4) **Learn from geometry-derived supervision.** The baseline is a non-learned reference, not a deployment target. Using the registration pipeline as a teacher for pseudo-labels, or fine-tuning a foundation-model backbone on homography-derived correspondence, would retain the location-based advantage on pairs registration cannot answer.
- 5) **Evaluate the ticket lifecycle.** Define task-level metrics — duplicate rate and missed-new-defect rate — and evaluate segmentation, re-identification, and ticket assignment jointly in a closed-loop simulation of real inspection workflows.

Together, these steps turn the present within-visit reference into a deployable multi-visit re-identification component for operational ticket management.

IX. CONCLUSION

We built a dataset that makes crack re-identification testable: CRACKID, 140 photographs over 17 walls with 212 identities in the frozen snapshot, its annotation provenance (45% merged) stated rather than assumed, with the triage of unclaimed clicks complete (107/1023 naming no crack). All accuracy rows are point estimates with no wall-level intervals (19 matchable test identities on the prior snapshot); we claim

the pattern, not the digits. The benchmark’s main finding is negative and structural: on the decision a record requires, ten zero-shot matchers across four families buy between +0.000 and +0.037 over the 0.337 best-constant floor (F_1^* 0.337–0.374), with all nine new rows at or below the floor at the validation threshold ($F_1^{@v}$ 0.331–0.338). Ordering a gallery and deciding an identity are not the same capability. The difficulty is not crack appearance alone: a geometric reference baseline (prior snapshot) that registers on wall texture with cracks masked out and Chamfer-matches projected masks reaches DIR@FAR=0.1 of 0.768 against 0.384 for the best new crop method and 0.404 for the best prior context method while answering 48% of pairs, and a COVERAGE-ONLY control exploiting that alignment without examining a crack scores 0.000 on the open-set metric (prior 0.544; 0.558 on current labels with old registration outcomes). Operationally, the open-set advantage comes from an aligned pair supporting a *licensed negative*, positive evidence of absence, where a crop embedding can only report low similarity; whether that advantage is independent of homography-based annotation remains unresolved and must be read as provenance-conditioned. Two bounds remain addressable: a real second capture session turns the benchmark from viewpoint to cross-visit, and a second annotator plus a homography-free audit subset would turn the single-annotator triage into a measured agreement rate. The 48% coverage is also a deployment gap in its own right, not merely a scoring issue: smooth-plaster walls can be categorically unanswerable. Finally, the small single-seed synthetic revisit checkpoint reverses the headline ranking and open-set result in favour of OSNet+context, while the distinct HYBRID-SKEL pipeline retains only a narrow assignment- F_1 advantage; the present paper therefore establishes a promising within-visit geometric reference, not a solved re-identification component for a multi-visit ticket management backend.

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REFERENCES

- [1] S. Kulkarni, S. Singh, D. Balakrishnan, S. Sharma, S. Devunuri, and S. C. R. Koralapati, “CrackSeg9k: A collection and benchmark for crack segmentation datasets and frameworks,” *arXiv:2208.13054*, 2022.
- [2] C. Benz and V. Rodehorst, “OmniCrack30k: A benchmark for crack segmentation and the reasonable effectiveness of transfer learning,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2024.
- [3] K. Zhou, Y. Yang, A. Cavallaro, and T. Xiang, “Omni-scale feature learning for person re-identification,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 3702–3712.
- [4] Y. Qian, J. Barthelemy, E. Karupiah *et al.*, “Identifying re-identification challenges: Past, current and future trends,” *SN Computer Science*, vol. 5, p. 937, 2024.
- [5] Q. Zou, Y. Cao, Q. Li, Q. Mao, and S. Wang, “CrackTree: Automatic crack detection from pavement images,” *Pattern Recognition Letters*, vol. 33, no. 3, pp. 227–238, 2012.

- [6] Y. Liu, J. Yao, X. Lu, R. Xie, and L. Li, "DeepCrack: A deep hierarchical feature learning architecture for crack segmentation," *Neurocomputing*, vol. 338, pp. 139–153, 2019.
- [7] M. Eisenbach, R. Stricker, D. Seichter, K. Amende, K. Debes, M. Sesselmann, D. Ebersbach, U. Stoeckert, and H.-M. Gross, "How to get pavement distress detection ready for deep learning? a systematic approach," in *Int. Joint Conf. Neural Networks (IJCNN)*, 2017, pp. 2039–2047.
- [8] C. Benz and V. Rodehorst, "CrackStructures and CrackEnsembles: The power of multi-view for 2.5d crack detection," in *Proceedings of the Winter Conference on Applications of Computer Vision (WACV)*, 2025, pp. 5990–5999.
- [9] L. Parente, E. Falvo, C. Castagnetti, F. Grassi, F. Mancini, P. Rossi, and A. Capra, "Image-based monitoring of cracks: Effectiveness analysis of an open-source machine learning-assisted procedure," *Journal of Imaging*, vol. 8, no. 2, p. 22, 2022.
- [10] S. Harb, P. Achanccaray, M. Maboudi, and M. Gerke, "Multi-temporal crack segmentation in concrete structures using deep learning approaches," *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. X-G-2025, pp. 341–348, 2025.
- [11] X. Sun and P. Chang, "Robust perspective correction for real-world crack evolution tracking in image-based structural health monitoring," *arXiv:2506.22437*, 2025.
- [12] NEC Corporation, "Photographing guide device," US Patent 12,223,639 B2 (priority 2019), 2025.
- [13] S. Liao, Z. Mo, Y. Hu, and S. Z. Li, "Open-set person re-identification," *arXiv preprint arXiv:1408.0872*, 2014.
- [14] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *International Journal of Computer Vision*, vol. 60, no. 2, pp. 91–110, 2004.
- [15] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski, "ORB: An efficient alternative to SIFT or SURF," in *IEEE International Conference on Computer Vision (ICCV)*, 2011, pp. 2564–2571.
- [16] P.-E. Sarlin, D. DeTone, T. Malisiewicz, and A. Rabinovich, "SuperGlue: Learning feature matching with graph neural networks," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 4938–4947.
- [17] J. Sun, Z. Shen, Y. Wang, H. Bao, and X. Zhou, "LoFTR: Detector-free local feature matching with transformers," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 8922–8931.
- [18] A. Dosovitskiy, L. Beyer, A. Kolesnikov *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *International Conference on Learning Representations (ICLR)*, 2021.
- [19] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers and distillation through attention," in *International Conference on Machine Learning (ICML)*, 2021, pp. 10 347–10 357.
- [20] A. Radford, J. W. Kim, C. Hallacy *et al.*, "Learning transferable visual models from natural language supervision," in *International Conference on Machine Learning (ICML)*, 2021, pp. 8748–8763.
- [21] M. Oquab, T. Darcet, T. Moutakanni *et al.*, "DINOv2: Learning robust visual features without supervision," *Transactions on Machine Learning Research*, 2024.
- [22] P. Lindenberger, P.-E. Sarlin, and M. Pollefeys, "LightGlue: Local feature matching at light speed," in *IEEE/CVF Int. Conference on Computer Vision (ICCV)*, 2023, pp. 17 627–17 638.
- [23] R. Arandjelović, P. Gronat, A. Torii, T. Pajdla, and J. Sivic, "NetVLAD: CNN architecture for weakly supervised place recognition," in *IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 5297–5307.
- [24] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," <https://github.com/ultralytics/ultralytics>, 2023.
- [25] L. Maiano, A. Montuschi, M. Caserio *et al.*, "A deep-learning-based antifraud system for car-insurance claims," *Expert Systems with Applications*, vol. 231, p. 120644, 2023.
- [26] M. J. Hasan, C. K. Nguyen, Y. L. Boo, H. Jahani, and K.-L. Ong, "Vehicle damage detection using artificial intelligence: A systematic literature review," *WIREs Data Mining and Knowledge Discovery*, vol. 15, p. e70027, 2025.
- [27] S. A. Pérez-Zarate, D. Corzo-García, J. L. Pro-Martín *et al.*, "Automated car damage assessment using computer vision: Insurance company use case," *Applied Sciences*, vol. 14, no. 20, p. 9560, 2024.
- [28] D. Mallios, L. Xiaofei, N. McLaughlin, J. M. Del Rincon, C. Galbraith, and R. Garland, "Vehicle damage severity estimation for insurance operations using in-the-wild mobile images," *IEEE Access*, vol. 11, pp. 78 644–78 655, 2023.
- [29] K. Sakurada and T. Okatani, "Change detection from a street image pair using CNN features and superpixel segmentation," in *British Machine Vision Conference (BMVC)*, 2015.
- [30] P. F. Alcantarilla, S. Stent, G. Ros, R. Arroyo, and R. Gherardi, "Street-view change detection with deconvolutional networks," *Autonomous Robots*, vol. 42, no. 7, pp. 1301–1322, 2018.
- [31] R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge University Press, 2004.
- [32] M. A. Fischler and R. C. Bolles, "Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography," in *Readings in Computer Vision*. Morgan Kaufmann, 1987, pp. 726–740.
- [33] D. Barath, J. Matas, and J. Noskova, "MAGSAC: Marginalizing sample consensus," in *IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 10 197–10 205.
- [34] G. D. Evangelidis and E. Z. Psarakis, "Parametric image alignment using enhanced correlation coefficient maximization," *IEEE Trans. Pattern Analysis and Machine Intelligence*, vol. 30, no. 10, pp. 1858–1865, 2008.
- [35] H. W. Kuhn, "The Hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, vol. 2, no. 1-2, pp. 83–97, 1955.